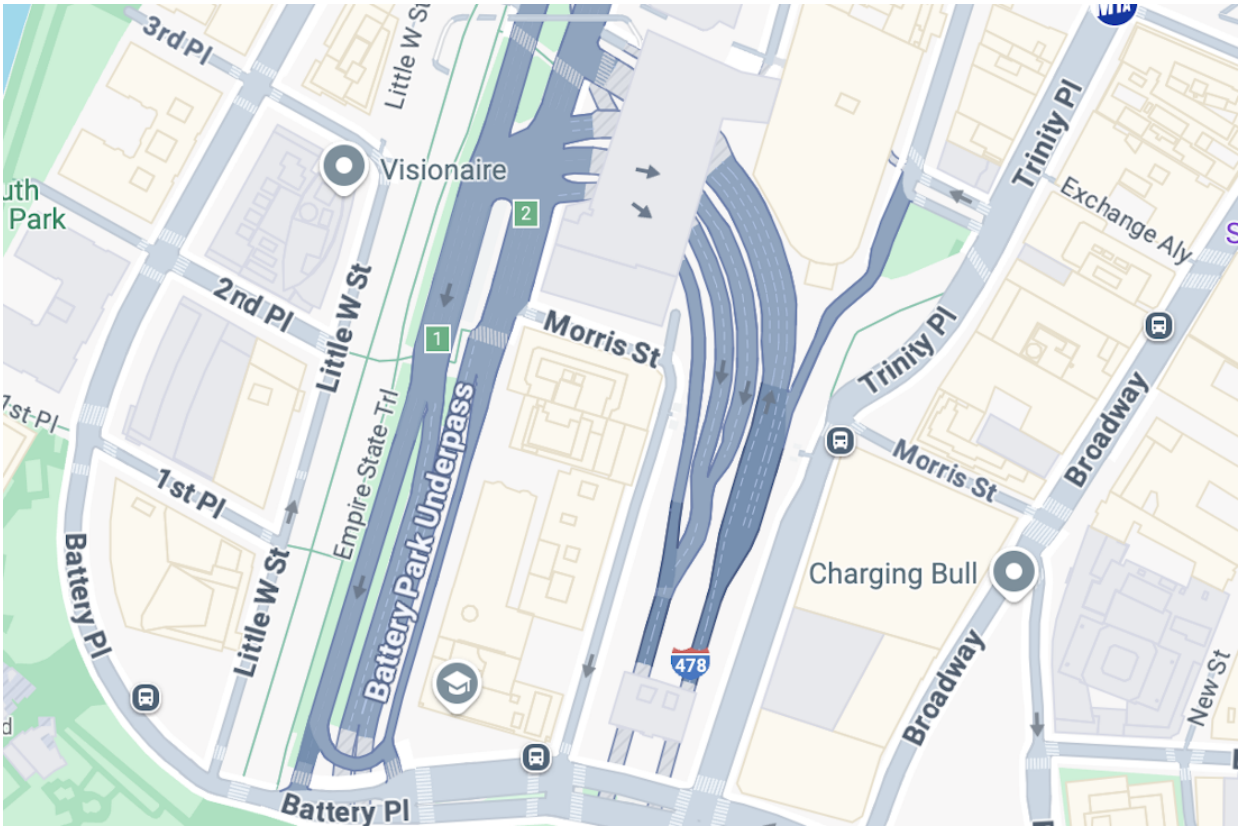


Figure 1.3 Google Maps view of lower Manhattan, a familiar example of a descriptive digital twin. Map data © 2025 Google. Google Maps is a trademark of Google LLC.

In recent years, Google Maps has evolved from simple two-dimensional maps to rich, photorealistic 3D renderings of the built environment, as shown in figure 1.4. These visuals include buildings modeled through photogrammetry, allowing users to "fly through" cities from the other side of the world, almost as if they were there in person. As one of the largest efforts ever undertaken to capture and represent the physical environment digitally, it is not only a descriptive digital twin itself, but also provides an unprecedented digital foundation for others to build spatially accurate digital twins on top of.